



APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

(A State Government University)

CURRICULUM, B. Tech 2024

**MBA Block, College of Engineering, Thiruvananthapuram Campus
Thiruvananthapuram- 695016**

1. Grouping

APJ Abdul Kalam Technological University offers various engineering branches that can be grouped into four broad categories based on their specialization.

Group	Branches
A Computer and Information Science	Computer Science and Engineering, Artificial Intelligence, CSE(AI), AI and Machine Learning, CSE (AI & ML), CS and Business Systems, CSE and BS, CS and Design, Cyber Security, AI and Data Science, Information Technology, CSE(DS),CSE(AI&DS), CSE (Internet of Things), CSE(IoT), CSE(Block Chain), CSE (Cyber Security), CSE(IoT and CS including Block Chain Technology), Cyber Physical System.
B Electrical Science	Electronics & Communication Engineering (ECE), Electrical and Electronics Engineering (EEE), Electronics & Instrumentation Engineering, Instrumentation and Control Engineering, Applied Electronics & Instrumentation Engineering, Electronics And Biomedical Engineering, Biomedical Engineering, Electronics and Computer Engineering, Electrical and Computer Engineering, Robotics and Artificial Intelligence, Robotics and Automation, B. Tech Electronics Engineering (VLSI Design and Technology), Electronics and Communication (Advanced Communication Technology).
C Physical Science	Civil Engineering, Chemical Engineering, Civil and Environmental Engineering, Mechanical Engineering, Mechanical Engineering (Auto), Mechanical Engineering (Automobile), Naval Architecture & Ship Building Engineering, Aeronautical Engineering, Industrial Engineering, Mechatronics Engineering, Metallurgical & Materials Engineering, Production Engineering, Automobile Engineering, Safety and Fire Engineering, Polymer Engineering.
D Life Science and Agriculture Engineering	Biotechnology, Biotechnology and Biochemical Engineering, Agriculture Engineering, Food Technology.

2. Course Category

- **University Core (UC):** This is a compulsory set of courses for all B. Tech students which includes basic courses in Mathematics, Humanities and Computer Science.
- **University Elective (UE):** These are elective courses from a basket of courses in the Humanities and Social Sciences. Such a UE course cannot be either UC / PC specified in their curriculum.
- **Group Core (GC):** Courses listed under Group Core of a curriculum are group specific. Students have to complete all the courses listed under GC to become eligible for the degree

FIRST SEMESTER (July-December): Group A and Group B														
10 Days Compulsory Induction Program														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./Week
						L	T	P	R		CIA	ESE		
1	A	GYMAT101	BSC	GC	Group Specific Mathematics -1	3	1	0	0	6	40	60	4	4
2	B S1/ S2	GYPHT121	BSC	GC	Physics for Engineers	2	0	2	0	5	40	60	3	4
		GYCYT122			Chemistry for Engineers	2	0	2	0	5	40	60		
3	C	GYEST103	ESC	GC	Engineering Graphics and Computer Aided Drawing.	2	0	2	0	5	40	60	3	4
4	D	GYEST104	ESC	GC	Introduction to Electrical & Electronics Engineering (part 1: Electrical Engineering)	2	0	0	0	4	20	30	2+2=4	4
					(Part 2: Electronics Engineering)	2	0	0	0	4	20	30		
5	F	UCEST105	ESC	UC	Algorithmic Thinking with Python	2	1	2	0	5	40	60	4	5
6	L	GYESL106	ESC	GC	Basic Electrical and Electronics Engineering Workshop	0	0	2	0	1	50	---	1	2
7	I* S1/ S2	UCPST127	PS	UC	Health and Safety	1	0	1	0	0	0	0	1	2
		UCHUT128	HMC		Life Skills and Universal Human Values	2	0	-	0	2	50	50		
8	S1/ S2	UCSEM129	SEC	UC	Skill Enhancement Course: Digital 101(30 Hours, NASSCOM)	MOOC				2			-	
Total										32/ 34			20	25
Bridge Course (Mathematics or Introduction to Computer Science) *:										Total 15 Hrs.				

*Valuation for HMC courses will be done at college level, Question papers will be provided by the University.

*No Grade Points will be awarded for the MOOC course and I slot course.

- L-T-P-R: Lecture-Tutorial-Practical-Project
- SS (Self Study) Hours= 2L+0.5P+R
- CIA: Continuous Internal Assessment, ESE: End Semester Examination

Digital 101 (NASSCOM)		
Sl. No:	Technologies Covered	Hours
1	Artificial intelligence and Big Data Analytics (AI/BDA)	11
2	Internet of Things (IoT)	2.5
3	Cyber Security	2.5
4	Block Chain	2.5
5	Robotic Process Automation	1.5
6	Augmented and Virtual Reality (AR and VR)	2.5
7	Cloud Computing	2.5
8	3 D Printing and Modelling	2
9	Web, Mobile Dev and Marketing	2
10	Responsible AI	1
Total Hours		30

Note: Engineering Physics, Engineering Chemistry, Health and Safety and Life skill and Universal Human Values shall be offered in both S1 and S2. Institutions can advise students belonging to about 50% of the number of branches in the Institution to opt for Engineering Physics/ Health and Safety in S1 and Engineering Chemistry/ Life Skills and UHV in S2 & vice versa.

FIRST SEMESTER (July-December): Group C														
10 Days Compulsory Induction Program														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./Week
						L	T	P	R		CIA	ESE		
1	A	GCMAT101	BSC	GC	Group Specific Mathematics -1	3	1	0	0	6	40	60	4	4
2	B S1/ S2	GCPHT121	BSC	GC	Physics for Engineers	2	0	2	0	5	40	60	3	4
		GCCYT122			Chemistry for Engineers	2	0	2	0	5	40	60		
3	C	GCEST103	ESC	GC	Engineering Mechanics	2	1	0	0	4	40	60	3	3
4	D	GCEST104	ESC	GC	Introduction to Mechanical Engineering & Civil Engineering (Part1: Mechanical Engineering)	2	0	0	0	4	20	30	2+2=4	4
					(Part 2: Civil Engineering)	2	0	0	0	4	20	30		
5	F	UCEST105	ESC	UC	Algorithmic Thinking with Python	2	1	2	0	5	40	60	4	5
6	L	GCESL106	ESC	GC	Engineering Workshop	0	0	2	0	1	50	---	1	2
7	I* S1/ S2	UCPST127	PS	UC	Health and Safety	1	0	1	0	0	0	0	1	2
		UCHUT128	HMC		Life Skills and Universal Human Values	2	0	-	0	2	50	50		
8	S1/ S2	UCSEM129	SEC	UC	Skill Enhancement Course: Digital 101(30 Hours, NASSCOM)	MOOC				2			-	
Total										31/ 33			20	24
Bridge Course (Mathematics or Introduction to Computer Science) *: Total 15 Hrs.														

**No Grade Points will be awarded for the MOOC course and I slot course.*

FIRST SEMESTER (July-December): Group D														
10 Days Compulsory Induction Program														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./Week
						L	T	P	R		CIA	ESE		
1	A	GDMAT101	BSC	GC	Group Specific Mathematics -1	3	1	0	0	6	40	60	4	4
2	B S1/ S2	GDPHT121	BSC	GC	Physics for Engineers	2	0	2	0	5	40	60	3	4
		GDCYT122			Chemistry for Engineers	2	0	2	0	5	40	60		
3	C	GDEST103	ESC	GC	Engineering Graphics and Computer Aided Drawing.	2	0	2	0	5	40	60	3	4
4	D	GD XX T104	ESC	GC	Introduction to Biotechnology/Food Technology/Agriculture Engineering	4	0	0	0	8	40	60	4	4
5	F	UCEST105	ESC	UC	Algorithmic Thinking with Python	2	1	2	0	5	40	60	4	5
6	L	GD XX L106	ESC	GC	Foundations of Biotechnology/Food Technology/Agriculture Engineering Lab	0	0	2	0	1	50	---	1	2
7	I* S1/ S2	UCPST127	PS	UC	Health and Safety	1	0	1	0	0	0	0	1	2
		UCHUT128	HMC		Life Skills and Universal Human Values	2	0	-	0	2	50	50		
8	S1/ S2	UCSEM129	SEC	UC	Skill Enhancement Course: Digital 101(30 Hours, NASSCOM)	MOOC				2			-	
Total										32/ 34			20	25
Bridge Course (Mathematics or Introduction to Computer Science) *:										Total 15 Hrs.				

**No Grade Points will be awarded for the MOOC course and I slot course.*

SECOND SEMESTER (January-June): Group A and Group B														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./Week
						L	T	P	R		CIA	ESE		
1	A	GYMAT201	BSC	GC	Group Specific Mathematics-2	3	1	0	0	6	40	60	4	4
2	B S1/ S2	GYPHT121	BSC	GC	Physics for Engineers	2	0	2	0	5	40	60	3	4
		GYCYT122			Chemistry for Engineers	2	0	2	0	5	40	60		
3	C	GYEST203	ESC	GC	Foundations of Computing: From Hardware Essentials to Web Design	2	1	0	0	4	40	60	3	3
4	D	GYEST204	ESC	GC	Programming in C	3	0	2	0	7	40	60	4	5
5	E	PCXXT205	PC	PC	Core-1	3	1	0	0	6	40	60	4	4
6	F	UCEST206	ESC	UC	Engineering Entrepreneurship & IPR	2	1	0	0	4	40	60	3	3
7	I* S1/ S2	UCPST127	PS	UC	Health and Safety	1	0	1	0	0	0	0	1	2
		UCHUT128	HMC		Life Skills and Universal Human Values	2	0	-	0	2	50	50		
8	L	GYESL208	ESC	GC	IT Workshop	0	0	2	0	1	50	---	1	2
9	S1/ S2	UCSEM129	SEC	UC	Skill Enhancement Course: Digital 101(30 Hours, NASSCOM)	MOOC							1	
Total										35			24	27

**No Grade Points will be awarded for the MOOC course and I slot course.*

SECOND SEMESTER (January-June): Group C														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./Week
						L	T	P	R		CIA	ESE		
1	A	GCMAT201	BSC	GC	Group Specific Mathematics-2	3	1	0	0	6	40	60	4	4
2	B S1/ S2	GCPHT121	BSC	GC	Physics for Engineers	2	0	2	0	5	40	60	3	4
		GCCYT122			Chemistry for Engineers	2	0	2	0	5	40	60		
3	C	GCEST203	ESC	GC	Engineering Graphics and Computer Aided Drawing	2	0	2	0	5	40	60	3	4
4	D	GCEST204	ESC	GC	Basic Electrical & Electronics Engineering (Part 1: Electrical Engineering)	2	0	0	0	4	20	30	2+2=4	4
					(Part 2: Electronics Engineering)	2	0	0	0	4	20	30		
5	E	PCXXT205	PC	PC	Core-1	3	1	0	0	6	40	60	4	4
6	F	UCEST206	ESC	UC	Engineering Entrepreneurship & IPR	2	1	0	0	4	40	60	3	3
7	I* S1/ S2	UCPST127	PS	UC	Health and Safety	1	0	1	0	0	0	0	1	2
		UCHUT128	HMC		Life Skills and Universal Human Values	2	0	-	0	2	50	50		
8	L	GCESL208	ESC	GC	Basic Electrical and Electronics Engineering workshop	0	0	2	0	1	50	---	1	2
9	S1/ S2	UCSEM129	SEC	UC	Skill Enhancement Course: Digital 101(30 Hours, NASSCOM)	MOOC							1	
Total										37			24	27

**No Grade Points will be awarded for the MOOC course and I slot course.*

SECOND SEMESTER (January-June)): Group D														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./Week
						L	T	P	R		CIA	ESE		
1	A	GDMAT201	BSC	GC	Group Specific Mathematics-2	3	1	0	0	6	40	60	4	4
2	B 1/2	GDPHT121	BSC	GC	Physics for Engineers	2	0	2	0	5	40	60	3	4
		GDCYT122			Chemistry for Engineers	2	0	2	0	5	40	60		
3	C	GDEST203	ESC	GC	Basic Mechanical & Civil Engineering	3	0	0	0	6	40	60	3	3
4	D	GDEST204	ESC	GC	Basic Electrical & Electronics Engineering (Part 1: Electrical Engineering)	2	0	0	0	4	20	30	2+2=4	4
					(Part 2: Electronics Engineering)	2	0	0	0	4	20	30		
5	E	PCXXT205	PC	PC	Core-1	3	1	0	0	6	40	60	4	4
6	F	UCEST206	ESC	UC	Engineering Entrepreneurship & IPR	2	1	0	0	4	40	60	3	3
7	I*	UCPST127	PS	UC	Health and Safety	1	0	1	0	0	0	0	1	2
		UCHUT128	HMC		Life Skills and Universal Human Values	2	0	-	0	2	50	50		
8	L	GDESL208	ESC	GC	Basic Electrical and Electronics Engineering Workshop	0	0	2	0	1	50	---	1	2
9	S ₁ / S ₂	UCSEM129	SEC	UC	Skill Enhancement Course: Digital 101(30 Hours, NASSCOM)	MOOC							1	
Total										38			24	26

**No Grade Points will be awarded for the MOOC course and I slot course.*

THIRD SEMESTER (July-December)														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./ Week
						L	T	P	R		CIA	ESE		
1	A	GYMAT301	BSC	GC	Group Specific Mathematics-3	3	1	0	0	6	40	60	4	4
2	B	PCXXT302	PC	PC	Core-2	3	1	0	0	6	40	60	4	4
3	D	PBXXT303	PC-PBL	PB	Core-PBL-1	3	0	0	1	7	60	40	4	4
4	F	GYEST304	ESC-PBL	GC	Group A: Digital Electronics & Logic Design Group B, C and D: Introduction to Artificial Intelligence and Data Science	2	1	0	1	4	60	40	4	4
5	G S3/S4	UCHUT345	HMC	UC	Engineering Economics and Financial Management	2	0	0	0	4	50	50	2	2
		UCHUT346			Engineering Ethics and Social Responsibility									
6	I*	UCHMT306	HMC	UC	Professional Communication and Soft Skills for Engineers	2	0	-	0	2	50	50	1	2
7	L	PCXXL307	PCL	PC	LAB-1	0	0	3	0	1.5	50	50	2	3
8	Q	PCXXL308	PCL	PC	LAB-2	0	0	3	0	1.5	50	50	2	3
9	R/M		VAC		REMEDIAL/MINOR/COURSE	3	1	0	0	6			4*	4*
Total										32/38			23/27*	26/30*

**No Grade Points will be awarded for the I slot course.*

FOURTH SEMESTER (January-June)														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./ Week
						L	T	P	R		CIA	ESE		
1	A	GYMAT401	BSC	GC	Group Specific Mathematics-4	3	1	0	0	6	40	60	4	4
2	B	PCXXT402	PC	PC	Core-3	3	1	0	0	6	40	60	4	4
3	C	PCXXT403	PC	PC	Core-4	3	1	0	0	6	40	60	4	4
4	D	PBXXT404	PC-PBL	PB	Core-PBL-2	3	0	0	1	7	60	40	4	4
5	E	PEXXT41N	PE	PE	PE-1	2	1	0	0	4	40	60	3	3
6	G S3/S4	UCHUT345	HMC	UC	Engineering Economics and Financial Management	2	0	0	0	4	50	50	2	2
		UCHUT346			Engineering Ethics and Social Responsibility									
7	L	PCXXL407	PCL	PC	LAB-3	0	0	3	0	1.5	50	50	2	3
8	Q	PCXXL408	PCL	PC	LAB-4	0	0	3	0	1.5	50	50	2	3
9	R/M /H		VAC		Remedial/Minor/Honours Course	3	1	0	0	6			4*	4*
Total										36/ 42			25/ 29*	27/ 31*

Note: Engineering Economics & Financial Management and Engineering Ethics shall be offered in both S3 and S4. Institutions can advise students belonging to about 50% of the number of branches in the Institution to opt for Engineering Economics & Financial Management in S3 and Engineering Ethics & Social Responsibility in S4 and vice versa.

FIFTH SEMESTER (July-December)														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs./ Week
						L	T	P	R		CIA	ESE		
1	A	PCXXT501	PC	PC	Core-5	3	1	0	0	6	40	60	4	4
2	B	PCXXT502	PC	PC	Core-6	3	1	0	0	6	40	60	4	4
3	C	PEXXT52N	PE	PE	PE-2	2	1	0	0	4	40	60	3	3
4	D	PBXXT504	PC-PBL	PB	Core-PBL-3	3	0	0	1	7	60	40	4	4
5	F	GYEST505	ESC	GC	Design Thinking and Product Development (Group Specific Syllabus)	2	1	0	0	4	40	60	3	3
6	I*	UCHUM506	HMC	UC	Constitution Of India (MOOC)	-	-	-	-	2	-	-	1	-
7	L	PCXXL507	PCL	PC	LAB-5	0	0	3	0	1.5	50	50	2	3
8	Q	PCXXL508	PCL	PC	LAB-6	0	0	3	0	1.5	50	50	2	3
9	R/M /H		VAC		Remedial/Minor/Honours Course, Mini Project (6 th Semester) Topic identification.	3	1	0	0	6			4*	4*
	S ₅ / S ₆	Industrial Visit (Maximum 10 Days are permitted, Not Exceeding more than 5 Working Days) /Industrial Training												
Total										32 / 38			23/27*	24/28*

*No Grade Points will be awarded for the MOOC course and I slot course.

SIXTH SEMESTER (January-June)														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs/ Week
						L	T	P	R		CIA	ESE		
1	A	PEXXT63N	PE	PE	PE-3	2	1	0	0	4	40	60	3	3
2	B	PEXXT64N	PE	PE	PE-4	2	1	0	0	4	40	60	3	3
3	C	PBXXT603	PC-PBL	PB	Core-PBL-4	3	0	0	1	7	60	40	4	4
4	D	PBXXT604	PC-PBL	PB	Core-PBL-5	3	0	0	1	7	60	40	4	4
5	O	OEXXT61N /IEXXT61N	OE/ILE	OE/IE	OE/ILE-1	2	1	0	0	4	40	60	3	3
6	I*	UCHUT606	HMC	UC	Sustainable Engineering	2	0	0	0	0	50	50	1	2
7	L	PCXXL607	PCL	PC	LAB-7	0	0	3	0	1.5	50	50	2	3
8	P	PCXXP608	PS	PC	Mini Project: Socially Relevant Project	0	0	0	3	3	50	50	2	3
	R/ M/ H		VAC		Remedial/Minor/Honours Course	3	1	0	0	6			4*	4*
	S5 / S6	Industrial Visit (Maximum 10 Days are permitted, Not Exceeding more than 5 Working Days) /Industrial Training												
Total										31/ 37			22/26*	25/29*

SEVENTH SEMESTER (July-December)														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs/ Week
						L	T	P	R		CIA	ESE		
1	A	PEXXT75N / PEXXM75N	PE	PE	PE-5 (Internship Students: Self Study/MOOC Approved by the University/Online Classes)	2	1	0	0	4	40	60	3	3
2	B	PEXXT76N/ PEXXM76N	PE	PE	PE-6 (Internship Students: Self Study/MOOC Approved by the University/Online Classes)	2	1	0	0	4	40	60	3	3
3	O	OEXXT72N /IEXXT72N/ OEXXM72N	OE/ ILE	OE/IE	OE/ILE-2 (Internship Students: Self Study/MOOC Approved by the University/Online Classes)	2	1	0	0	4	40	60	3	3
4	I*	UEHUT704 / UEHUM70N	HMC	UE	Elective (Internship Students: Self Study/MOOC Approved by the University/Online Classes)	2	0	0	0	4	50	50	2	2
5	S	PCXXS705	PS	PC	Seminar	0	0	3	0	1.5	50	0	2	3
6	P	PCXXP706/ PCXXI706	PS	PC	Option 1: Major Project Option 2: Internship (4-6 Months)	0	0	0	14	14	100	0	5	14
	R/M /H		VAC		Remedial/Minor/Honours Course	0	0	0	4	4			4*	4*
Total										31/ 35			18/23*	28/32*

Slot I: HMC Elective	
1	Project Management: Planning, Execution, Evaluation and Control
2	Proficiency course in French . (MOOC) (B1 level)
3	Proficiency Course in German (B1 Level) . (MOOC)
4	Proficiency Course in Spanish (B1 Level) (MOOC)
5	Introduction to Japanese Language and Culture (N5 level) . (MOOC)

**No Grade Points will be awarded for the I slot courses*

EIGHT SEMESTER (January-June)														
Sl. No:	Slot	Course Code	Course Type	Course Category	Course Title (Course Name)	Credit Structure				SS	Total Marks		Credits	Hrs/Week
						L	T	P	R		CIA	ESE		
1	A	PEXXT87N / PEXXM87N	PE	PE	PE-7 (Internship Students: Self Study/MOOC Approved by the University/Online Classes)	2	1	0	0	4	40	60	3	3
2	O	OEXT82N /IEXT82N / OEXXM82N	OE/ILE	OE/IE	OE/ILE-3 (Internship Students: Self Study/MOOC Approved by the University/Online Classes)	2	1	0	0	4	40	60	3	3
3	I*	UEHUT803 / UEHUM803	HMC	UC	Organizational Behavior and Business Communication (Internship Students: Self Study/MOOC Approved by the University/Online Classes)	2	0	0	0	2	50	50	1	2
4	P	PCXXP806/ PCXXI806/ PCXXJ806	PS	PC	Option 1: Major Project Option 2: Internship (4-6 Months) Option 3: Major Project Phase -II (For the students who have not opted for internship in S7/S8)	0	0	0	14	14	100	0	5	14
	R/H		VAC		Project: Honours Course	0	0	0	4	4			4*	4
Total										24 / 28			12/16*	22/26

***No Grade Points will be awarded for the I slot courses**

Note: Open Electives are such courses which will be offered by other departments. Like CSE department students have to opt open electives from ECE/ME/EEE etc. departments.

Dr. Libish T M
Director (Academic)
APJ Abdul Kalam Technological University

Dr. Vinu Thomas
Dean (Academic)
APJ Abdul Kalam Technological University

HMC Courses				
Sl. No:	Semester	Course Code	Course Area	Credits
1	S1/S2		Life Skills and Universal Human Values	1
2	S3		Professional Communication and Soft Skills for Engineers	1
3	S3		Engineering Economics and Financial Management	2
4	/S4		Engineering Ethics and Social Responsibility	2
5	S5		Constitution Of India.(MOOC)	1
6	S6		Sustainable Engineering	1
7	S7		Elective (Project Management/Foreign Languages)	2
8	S8		Organizational Behavior and Business Communication	1
Total Credits				11

BSC Courses				
Sl. No:	Semester	Course Code	Course Area	Credits
1	S1		Group Specific Mathematics-1	4
2	S1/S2		Physics for Engineers	3
3			Chemistry for Engineers	3
4	S2		Group Specific Mathematics-2	4
5	S3		Group Specific Mathematics-3	4
6	S4		Group Specific Mathematics-4	4
Total Credits				22

ESC Courses (Group A and B)			
Sl. No:	Semester	Course Area	Credits
1	S1	Engineering Graphics and Computer Aided Drawing	3
2		Introduction to Electrical and Electronics Engineering	4
3		Algorithmic Thinking with Python	4
4		Basic Electrical and Electronics Engineering Workshop	1
5	S2	Foundations of Computing: From Hardware Essentials to Web Design	3
6		Programming in C	4
7		Engineering Entrepreneurship and IPR	3
8		IT Workshop	1
9	S3	Group1: Digital Electronics & Logic Design Group2: Introduction to Artificial Intelligence and Data Science	4
10	S5	Design Thinking and Creativity	3
Total Credits			30

ESC Courses (Group C)			
Sl. No:	Semester	Course Area	Credits
1	S1	Engineering Mechanics	3
2		Introduction to Mechanical Engineering/ Civil Engineering	4
3		Algorithmic Thinking with Python	4
4		Engineering Workshop	1
5	S2	Engineering Graphics and Computer Aided Drawing	3
6		Basic Electrical and Electronics Engineering	4
7		Engineering Entrepreneurship and IPR	3
8		Basic Electrical and Electronics Engineering Workshop	1
9	S3	Introduction to Artificial Intelligence and Data Science	4
10	S5	Design Thinking and Creativity	3
Total Credits			30

ESC Courses (Group D)			
Sl. No:	Semester	Course Area	Credits
1	S1	Engineering Graphics and Computer Aided Drawing	3
2		Introduction to Biotechnology/Food Technology/Agriculture Engineering	4
3		Algorithmic Thinking with Python	4
4		Foundations of Biotechnology/Food Technology/Agriculture Engineering Lab	1
5	S2	Basic Mechanical Engineering and Civil Engineering	3
6		Basic Electrical and Electronics Engineering	4
7		Engineering Entrepreneurship and IPR	3
8		Basic Electrical and Electronics Engineering Workshop	1
9	S3	Introduction to Artificial Intelligence and Data Science	4
7	S5	Design Thinking and Creativity	3
Total Credits			30

Programme Core Courses(PC)			
Sl. No:	Semester	Course Area	Credits
1	S2	Core 1	4
2	S3	Core 2	4
3		Lab-1	2
4		Lab-2	2
5	S4	Core 3	4
6		Core 4	4
7		Lab-3	2
8		Lab-4	2
9	S5	Core 5	4
10		Core 6	4
11		Lab-5	2
12		Lab-6	2
13	S6	Lab-7	2
Total Credits (Theory -6, Lab-7)			38

Programme Core-Project Based Learning (PBL)			
Sl. No:	Semester	Course Area	Credits
1	S3	Core PBL-1	4
2	S4	Core PBL-2	4
3	S5	Core PBL-3	4
4	S6	Core PBL-4	4
5		Core PBL-5	4
Total Credits			20

Programme Elective Courses (PE)			
Sl. No:	Semester	Course Type	Credits
1	S4	PE-1	3
2	S5	PE-2	3
3	S6	PE-3	3
4		PE-4	3
5	S7	PE-5	3
6		PE-6	3
7	S8	PE-7	3
Total Credits			21

Open Elective Courses/Industry Elective(OE/IEL)			
Sl. No:	Semester	Course Type	Credits
1	S6	OE/ILE-1	3
2	S7	OE/ILE-2	3
3	S8	OE/ILE-3	3
Total Credits			9

Project/Seminar			
Sl. No:	Semester	Course Type	Credits
1	S6	Mini Project: Socially Relevant Project	2
2	S7	SEMINAR	2
3		MAJOR PROJECT/Internship	5
4	S8	MAJOR PROJECT/Internship/Research Project	5
Total Credits			14

Sl. No	Group	Courses	Credits	Minimum Credit Requirements
1	I	NSS, NCC, NSO (National Sports Organization)	1 (40 Points)	3 Credits (One credit from each Group)
2		Arts/Sports/Games		
3		Union/Club Activities		
4	II	English Proficiency Certification (TOFEL, IELTS, BEC etc)	1 (40 Points)	
5		Aptitude Proficiency Certification (GRE, CAT, GMAT etc)/Valid Gate Score		
6		Short Term Internship (Minimum 4 weeks), Conference/Workshop Activities/ Professional Body Activities.		
7	III		1 (40 Points)	
8		Journal Publication		
9		Skilling Certificates (Approved by the University)		

***30 Points/group for B. Tech Lateral Entry Students**

- ***A minimum of 120 Activity points is to be acquired for obtaining the 3 Activity Credits required in the curriculum***